

MUSCATINE, IA, USA

WMO: 725487

Lat: 41.367N

Lon: 91.150W

Elev: 547

StdP: 14.41

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04950

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.4	(c) 2.7	(d) -11.4	(e) 3.0	(f) -0.6	(g) -7.7	(h) 3.7	(i) 4.2	(j) 29.1	(k) 25.8	(l) 26.1	(m) 23.8	(n) 7.5	(o) 300	(p) 0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	94.8	77.3	91.1	75.8	89.6	74.8	80.6	90.0	78.7	88.1	77.0	85.7	9.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.5	151.2	86.5	76.3	140.1	84.5	73.4	126.9	82.0	44.6	90.4	42.6	88.0	40.6	86.3	94.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.9	(b) 21.0	(c) 18.7	(e) -13.0	(f) 98.6	(g) 10.5	(h) 4.1	(i) -20.5	(j) 101.5	(k) -26.6	(l) 103.9	(m) -32.5	(n) 106.2	(o) -40.1	(p) 109.1		
(a) 24.9	(b) 21.0	(c) 18.7	(e) -12.4	(f) 84.1	(g) 9.7	(h) 3.7	(i) -19.4	(j) 86.7	(k) -25.1	(l) 88.9	(m) -30.5	(n) 91.0	(o) -37.6	(p) 93.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.8	24.3	28.0	39.5	51.8	62.9	72.5	76.5	73.7	67.0	54.0	40.3	29.2
	DBStd	20.51	12.02	11.86	11.60	9.76	8.47	6.49	6.15	5.48	7.60	9.24	10.35	10.94
	HDD50	2910	796	619	363	92	6	0	0	0	0	65	321	648
	HDD65	5933	1260	1037	792	405	145	13	2	4	66	359	741	1109
	CDD50	3561	0	3	39	148	404	676	822	736	510	188	31	4
	CDD65	1109	0	0	2	11	79	239	359	275	126	17	1	0
	CDH74	11353	0	0	29	177	784	2367	3899	2554	1321	203	8	11
	CDH80	4514	0	0	4	36	234	955	1764	982	484	47	0	8
Wind	WSAvg	7.6	8.7	8.8	9.2	9.7	8.2	6.8	5.5	4.9	5.7	7.5	8.4	8.1
Precipitation	PrecAvg	38.4	1.4	1.7	2.7	3.7	5.0	5.6	4.2	4.4	3.1	2.8	2.0	1.8
	PrecMax	55.4	2.8	4.3	5.9	6.5	12.1	11.9	11.5	11.7	8.3	7.2	5.6	3.8
	PrecMin	21.0	0.3	0.4	0.6	1.2	0.9	1.6	0.6	0.9	0.6	0.5	0.2	0.5
	PrecStd	8.9	0.6	1.1	1.8	1.4	2.5	3.0	2.7	2.6	2.1	1.9	1.4	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.8	63.7	78.9	84.5	90.7	96.6	100.2	96.6	93.3	85.8	74.5	63.8
		MCWB	46.8	53.3	64.1	65.1	72.4	77.3	78.1	77.9	74.2	69.1	61.5	58.6
	2%	DB	49.7	55.1	70.2	80.5	85.8	92.5	95.4	91.3	90.1	79.4	66.5	54.8
		MCWB	44.7	47.7	58.1	63.4	69.4	74.9	79.1	77.2	72.9	63.6	56.4	49.9
	5%	DB	43.5	48.5	63.4	73.9	81.8	89.6	91.4	88.5	85.9	74.7	62.6	48.4
		MCWB	38.9	43.2	54.4	59.4	66.9	73.3	77.0	75.6	71.0	61.6	53.9	44.6
	10%	DB	38.9	43.3	57.1	69.8	78.7	85.8	89.8	85.9	81.7	70.2	56.7	44.6
		MCWB	35.8	38.7	48.9	57.0	65.2	71.1	76.1	74.0	68.1	59.8	49.2	40.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.9	55.5	65.6	69.2	75.9	80.5	84.0	83.1	78.2	72.4	63.5	60.3
		MCDB	52.3	60.0	75.4	80.3	86.1	91.8	93.2	90.9	88.3	81.5	72.7	64.0
	2%	WB	43.8	49.0	60.3	65.7	72.4	78.0	81.2	80.1	75.5	68.5	59.0	51.3
		MCDB	48.6	55.3	67.7	75.6	82.0	88.5	90.5	88.4	85.8	74.1	64.7	53.6
	5%	WB	39.6	43.3	55.0	62.3	70.1	75.5	79.3	78.0	73.5	64.6	54.8	45.3
		MCDB	43.9	48.8	61.4	71.4	78.7	85.3	89.6	85.5	81.5	70.7	61.0	48.2
	10%	WB	36.3	39.0	49.9	58.7	67.4	73.3	77.6	76.0	71.3	61.3	50.2	40.0
		MCDB	39.2	43.1	57.2	68.2	75.7	82.6	87.0	82.7	79.8	69.1	56.0	43.9
Mean Daily Temperature Range	5% DB	MDBR	15.7	16.5	18.9	21.7	20.7	20.5	20.8	20.7	23.8	21.6	18.4	15.0
		MCDBR	21.2	24.3	27.2	29.7	24.8	24.5	23.6	24.2	27.7	28.7	26.4	22.0
		MCWBR	16.5	17.3	16.9	16.7	12.4	11.0	10.9	11.9	12.9	15.4	17.6	17.6
	5% WB	MCDBR	20.6	23.6	24.2	24.7	21.7	21.4	21.0	20.3	22.4	21.1	24.5	20.4
		MCWBR	16.2	17.4	16.6	16.2	12.7	11.5	11.2	11.5	11.8	14.2	18.2	17.7
Clear-Sky Solar Irradiance	taub	0.292	0.307	0.342	0.388	0.430	0.428	0.445	0.428	0.396	0.348	0.305	0.293	
	taud	2.422	2.390	2.378	2.259	2.205	2.266	2.199	2.258	2.309	2.443	2.496	2.437	
	Ebn at Noon	274	287	287	279	269	270	263	265	266	268	268	264	
	Edh at Noon	25	30	34	42	45	43	45	41	36	29	23	23	
All-Sky Solar Radiation	RadAvg	571	883	1190	1493	1752	1957	1952	1720	1461	967	647	473	
	RadStd	67	79	97	123	126	184	133	107	116	109	54	55	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page